LIBXS Samples

Memory Comparison Benchmark

Benchmarks `libxs_diff`, `libxs_memcmp`, and the C standard library's `memcmp` for both large contiguous comparisons and many small fixed-size comparisons. A Fortran variant compares `libxs_diff` against Fortran's `ALL(a .EQ. b)` and `.NOT. ANY(a .NE. b)` array intrinsics.

Building

```
cd samples/memory
make
```

Produces `memory.x` (C) and, if a Fortran compiler is found, `memoryf.x` and `matcopyf.x`.

Usage (C)

```
./memory.x [elsize [stride [nelems [niters]]]]
```

Argument	Default	Description
`elsize`	`INSIZE`/`MAXSIZE` (compile-time, 64)	Element comparison size in bytes
`stride`	`max(MAXSIZE, elsize)`	Distance between successive elements in bytes
`nelems`	`2 GB / stride`	Number of elements
`niters`	5	Repetitions (arithmetic average is reported)

Environment Variables

Variable	Default	Description
`STRIDED`	0	**0**: `libxs_memcmp` compares the entire buffer in one call when `stride == elsize`, otherwise loops; **1**: sequential loop over elements; **>=2**: random-order traversal using a coprime-based permutation
`CHECK`	0	**Non-zero**: run a validation pass that injects random byte differences and cross-checks `libxs_diff` / `libxs_memcmp` against `memcmp`
`MISMATCH`	0	**0**: compare equal buffers (baseline); **1**: inject difference at first byte of each element; **2**: middle; **3**: last byte; **>=4**: random position. Tests early-out behavior of each implementation
`OFFSET`	0	Shift buffer `b` by this many bytes off its natural alignment. Tests SIMD alignment sensitivity

Compile-time Knobs

Macro	Default	Description
`MAXSIZE`	64	Stride floor; elements are spaced at least this far apart
`INSIZE`	`MAXSIZE`	Default element size when the first argument is omitted or zero

```
## 32-byte elements, sequential strided access, 10 repetitions
./memory.x 32 0 0 10

## 8-byte elements, random traversal, 3 repetitions
STRIDED=2 ./memory.x 8 0 0 3

## contiguous (single-call) comparison, ~2 GB
./memory.x 64 64 0 5

## mismatch at last byte, buffer b misaligned by 3 bytes
MISMATCH=3 OFFSET=3 ./memory.x 32 0 0 5
```

Usage (Fortran)

```
./memoryf.x [nelements [nrepeat]]
```

Element type is hard-coded as `INTEGER(4)`. Compares ~2 GB by default... reports per-iteration times (ms) and average throughput (MB/s).

What Is Measured

1. `libxs_diff` SIMD-dispatched short-buffer comparison (SSE/AVX2/AVX-512 depending on CPU). Limited to `elsize < 256` bytes. Always called per-element (strided).
2. `libxs_memcmp` SIMD-dispatched `memcmp` replacement. Used in a single call for contiguous buffers, or per-element for strided access.
3. **stdlib** `memcmp` The C library implementation, tested under the same access pattern.

Each kernel gets freshly initialized data (`libxs_rng_seq` + `memcpy`) before its timed section, so that caches are refilled from scratch and one kernel's residency does not gift the next one. An untimed warm-up pass runs before the first measured iteration to demand-page the buffers and stabilize CPU frequency.

When `MISMATCH` is set, a controlled byte difference is injected into buffer `b` at the configured position of each element after the `memcpy`. This allows measuring early-out behavior independently from the equal-buffer baseline.

The summary reports **min / median / max** across iterations plus peak MB/s (from the minimum time), giving a more robust picture than a simple arithmetic average.

The random traversal mode (`STRIDED>=2`) uses `libxs_coprime2(size)` to produce a full permutation of the element indices, stressing TLB/cache miss behavior.

Design Notes

Resolved Issues The following concerns from an earlier review have been addressed:

1. **Mismatch testing** (`MISMATCH` env var) Buffers can now differ at a configurable position (first byte, middle, last byte, or random), testing early-out performance in addition to the equal-buffer baseline.
2. **Min / median / max reporting** The summary section now shows the minimum, median, and maximum time across iterations, plus peak MB/s derived from the fastest run. This is more robust for memory-bandwidth benchmarks than an arithmetic mean.
3. **Warm-up iteration** An untimed pass through `libxs_rng_seq` + `memcpy` is run before the first measured iteration to demand-page the allocations and give the CPU time to ramp its frequency.
4. **Misalignment testing** (`OFFSET` env var) Buffer `b` can be shifted off its natural alignment to reveal SIMD alignment sensitivity.
5. `libxs_diff` **skip message** When `elsize >= 256`, the benchmark now prints a note instead of silently omitting the column.
6. **Deterministic validation** The `CHECK` mode now seeds `srand(42)` before the validation loop for reproducible results.
7. **POT element-size sweep** The test script (tests/memcmp.sh) now includes power-of-two sizes (1, 2, 4, 8, 16, 32, 64, 128) and non-POT neighbors (3, 5, 7, 9, 15, 31, 33, 63, 65, 127) to expose stdlib regressions on short fixed-size comparisons, which historically affected certain glibc versions.

Throughput Convention Reported MB/s counts both buffers (`2 * nbytes / time`) since both `a` and `b` are read during comparison. This differs from some published benchmarks that report single-buffer bandwidth.

Ozaki-Scheme Low-Precision GEMM

Intercepted GEMM

This code sample intercepts all four standard BLAS GEMM routines DGEMM, SGEMM, ZGEMM, and CGEMM and only relies on the LAPACK/BLAS interface. Real GEMM calls (DGEMM/SGEMM) are executed via one of three Ozaki low-precision schemes (mantissa slicing, CRT, or BF16 Dekker split); complex GEMM calls (ZGEMM/CGEMM) are implemented via the 3M (Karatsuba) method using three real GEMM calls each. The wrapper sources are compiled twice (once for `double`, once for `float`), so all four symbols coexist in a single binary.

Two link-time variants are built per precision: (1) code which is dynamically linked against LAPACK/BLAS (`dgemm-blas.x`/`sgemm-blas.x`), (2) code which is linked using `--wrap=`*symbol* supported by GNU GCC compatible tool chains (`dgemm-wrap.x`/`sgemm-wrap.x`). Running `test-wrap.sh` exercises three flavors: the two build variants and additionally the first variant using the LD_PRELOAD mechanism (available under Linux). The `test-check.sh` script validates all four Ozaki schemes for correctness, and `test-mhd.sh` tests MHD file-based GEMM input pairs.

The static wrapper library is built by default (`make`), and suitable for applications with static linkage against a LAPACK/BLAS library (`-Wl,--wrap=dgemm_ -Wl,--wrap=sgemm_ -Wl,--wrap=zgemm_ -Wl,--wrap=cgemm_`). To build and use the shared wrapper library:

```
cd /path/to/libxs
make -j $(nproc)

cd samples/ozaki
make BLAS_STATIC=0

LD_PRELOAD=/path/to/libwrap.so ./application
```

Note: LIBXS has to be built upfront for the sample code to link.

Performance

The Ozaki sample code demonstrates how to intercept GEMMs (any BLAS library), to run a low-precision GEMM instead of the original GEMM, to compare the results, and to maintain running statistics presented every Nth call of GEMM or when the application terminates. The code explores high-precision emulation using low-precision calculation.

The blocking structure is designed to conceptually emulate fixed-size matrix-multiply hardware. All computation operates on tiles of size `BLOCK_M` x `BLOCK_K` (A) and `BLOCK_K` x `BLOCK_N` (B), with defaults BLOCK_M = BLOCK_N = BLOCK_K = 1 The compile-time parameter `BATCH_K` (default 4) groups `BATCH_K` consecutive BLOCK_K panels into a single batch, so the effective K-dimension step per batch is BATCH_K x BLOCK_K. Batching reduces OpenMP barrier overhead and improves temporal reuse of the C tile, while keeping the fundamental tile size visible throughout the code.

Preprocessing (exponent alignment, mantissa slicing or modular reduction) accounts for roughly 5% of runtime; the remaining 95% is spent in the inner dot-product loops. In Scheme 1, int8 dot products are dispatched once per GEMM via a function pointer: AVX-512 VNNI (`VPDPBUSD`) when available, otherwise a scalar fallback. Scheme 3 uses BF16 Dekker splitting (no exponent alignment needed) and BF16 dot products (`VDPBF16PS`) instead of int8. The number of pairwise slice products is quadratic in the number of slices, so for double-precision (8 slices, default) the inner loop performs up to 36 dot products per block pair. Scheme 2 performs one modular dot product per prime, so its cost is linear in the number of primes.

OpenMP parallelizes all three phases of each K-batch: Phase 1 preprocesses A panels (`schedule(dynamic) nowait`), Phase 2 preprocesses B panels (`schedule(dynamic)`, implicit barrier), and Phase 3 accumulates the dot products into C (`collapse(2) schedule(static)`). Panel buffers are shared across the parallel region and sized by `BATCH_K` x number-of-blocks, allocated via `libxs_malloc`.

Practically all CPUs provide higher instruction throughput using floating point instructions even when relying on double-precision. The algorithmic complexity and inner-most code is in fact unsuitable to reach high performance levels. OpenMP based parallelization or VNNI instructions are only meant to improve emulating high-precision.

When built with LIBXSTREAM support (sibling `libxstream` repository detected at build time), an optional OpenCL/GPU path is available for Ozaki-1 and Ozaki-2 as well (see `OZAKI` environment variable). The GPU bridge (`ozaki_ocl.c`) wraps LIBXSTREAM behind an opaque handle so that the CPU code has no OpenCL dependency.

At runtime, the GPU path is enabled by default (`OZAKI_OCL=1`) and can be disabled with `OZAKI_OCL=0` to fall back to the CPU implementation. The CPU-only code performs the low-precision conversion on-the-fly and only requires a reasonable stack size to buffer small matrix blocks.

Scheme 1 Mantissa Slicing

Scheme 1 (`OZAKI=1`, the default) decomposes each IEEE-754 mantissa into 7-bit int8 slices and accumulates all pairwise slice products via low-precision GEMM. The number of slices aka "splits" determines achievable accuracy and can be set at runtime via `OZAKI_N`. The default and maximum vary by precision (double: default 8, max 16; float: default 4, max 8). The size of the matrices employed by potential "matrix cores" is set at compile-time with `BLOCK_M`, `BLOCK_N`, and `BLOCK_K`; grouping of consecutive K-panels is controlled by `BATCH_K`. The term "slices" is preferred over "splits" since the latter suggests *N* splits would yield *N+1* slices.

Complex GEMM (3M Method)

Intercepted ZGEMM (double-complex) and CGEMM (single-complex) calls are implemented via the Karatsuba 3M method. Each complex GEMM is decomposed into three real GEMM calls:

- P1 = Re(A) * Re(B)
- P2 = Im(A) * Im(B)
- P3 = (Re(A) + Im(A)) * (Re(B) + Im(B))

The real and imaginary parts of the product are recovered as Re(A*B) = P1 - P2 and Im(A*B) = P3 - P1 - P2. Complex alpha/beta scaling is applied in a final pass. The three real GEMM calls flow through the same wrapper, so they are optionally accelerated by the Ozaki scheme as well.

Scheme 1 Flags (`OZAKI_FLAGS`) and Diagonal Trim (`OZAKI_TRIM`)

The Scheme 1 loop is controlled by two runtime knobs:

Flags The environment variable `OZAKI_FLAGS` is an integer bitmask:

| Bit | Value | Flag | Description | |::|::|::|| 0 | 1 | Triangular | Only iterate slice pairs (sa, sb) with sb >= sa (upper triangle). | | 1 | 2 | Symmetrize | For each off-diagonal pair (sa, sb) in the upper triangle, also compute the mirror dot product D(sb, sa) at negligible extra cost (one additional int8 dot product per pair). |

The default value is **3** (both flags). Setting `OZAKI_FLAGS=0` runs the full S^2 square of slice pairs.

Diagonal Trim The environment variable `OZAKI_TRIM` drops the T least significant diagonals from the slice-pair iteration. Pairs with sa + sb > 2*(S-1) - T are skipped. Pair significance scales as 2^(low_bit[sa] + low_bit[sb]), so sa + sb directly determines significance smaller sums are more significant. Each dropped diagonal loses approximately 7 bits of relative precision.

The default value is **0** (exact: all pairs). The value is clamped so that at least diagonal 0 is always computed.

Cost overview for *S* slices with TRIANGULAR + SYMMETRIZE (default flags):

| `OZAKI_TRIM` | Dot products | Pairs covered | Relative bits lost | |::|::|::|::| 0 (default) | S*(S+1)/2 | all S^2 | 0 (exact) | | S-1 | ~$S^2/4$ | (S+1)*S/2* | ~7(S-1) least significant | | 3S/2 | ~$S^{2/8}$ | ~$S^2/4$ | *~73S/2 least significant* | | *2*(S-1) | 1 | 1 (only diagonal) | ~7*2*(S-1) |

Because SYMMETRIZE computes both D(sa,sb) and D(sb,sa), the number of dot products with TRIANGULAR equals S*(S+1)/2 for cutoff = 2*(S-1) but covers all S^2 contributions.

Examples:

```
./dgemm-wrap.x 256                    # exact (default: flags=3, trim=0)
OZAKI_TRIM=4 ./dgemm-wrap.x 256      # drop 4 least significant diagonals
OZAKI_FLAGS=0 ./dgemm-wrap.x 256     # full S^2 square, no symmetrize
```

Scheme 2 Chinese Remainder Theorem

Scheme 2 (`OZAKI=2`) uses modular arithmetic instead of mantissa slicing. Each matrix element is reduced modulo a set of small pairwise coprime moduli (primes and prime powers <= 128) so that residues fit in int8 and dot products use VNNI int8 instructions when available. GEMM is performed independently modulo each modulus, and the exact integer result is recovered via the Chinese Remainder Theorem (Garner's algorithm with grouped uint64 Horner evaluation). The Horner reconstruction partitions mixed-radix digits into groups of up to 9, evaluates each group exactly in uint64 arithmetic, and combines groups with a minimal number of FP64 operations avoiding double-precision throughput bottlenecks on hardware where integer arithmetic is faster. Because the work is linear in the number of moduli versus quadratic in the number of slices for Scheme 1 Scheme 2 can be more efficient when many moduli/slices are needed.

Residues are signed int8 (-127..+127) with the element sign folded directly into the residue. This maps naturally to VNNI's VPDPBUSD encoding (unsigned x signed with bias correction). An unsigned variant (uint8 residues, moduli <= 256) is theoretically possible but would require a separate sign array and scalar dot-product accumulation, negating the VNNI advantage.

The number of moduli can be set at runtime via `OZAKI_N`. The default and maximum are: double: 17 / 18; float: 8 / 10.

Example:

```
OZAKI=2 ./dgemm-wrap.x 256                          # use CRT scheme
```

Scheme 3 BF16 Dekker Split

Scheme 3 (`OZAKI=3`) decomposes each matrix element into a sequence of BF16 slices using a Dekker-style error-free split: `slice[0] = round_bf16(x); residual = x - slice[0]; slice[1] = round_bf16(residual); ...` Each BF16 value carries its own 8-bit exponent, so no shared per-row or per-column exponent is needed unlike Schemes 1 and 2, there is no exponent alignment, mantissa extraction, or `pow2` reconstruction. Preprocessing simply splits and scatters; accumulation is `C[i,j] += alpha * dot_bf16(A_sa, B_sb)` with no scale factors.

Each BF16 slice captures roughly 8 significant bits. For double precision (52 mantissa bits), ~7 slices suffice ($8\times7 = 56 > 52$); for single precision (23 bits), ~3 slices suffice. The dot products are computed via `VDPBF16PS` (AVX-512-BF16) when available, otherwise a scalar fallback. FP32 accumulation of BF16×BF16 products is **not** exact in general (unlike the integer-exact Scheme 1), so Scheme 3 is inherently approximate the error depends on the number of slices and the FP32 rounding during accumulation. In practice the relative error is very small (typically < 1e-9 for double precision with default settings).

Scheme 3 shares the same runtime knobs as Scheme 1: `OZAKI_N` (number of slices), `OZAKI_FLAGS` (triangular + symmetrize), and `OZAKI_TRIM` (diagonal trim). Because no per-row/per-column exponent panels are needed, the metadata overhead is halved relative to Scheme 1.

Example:

```
OZAKI=3 ./dgemm-wrap.x 256                          # use BF16 scheme
```

Compile-Time Parameters

The block and batch sizes can be overridden at compile time via `-D`:

Parameter	Default	Description
`BLOCK_M`	16	Tile rows (A and C).
`BLOCK_N`	16	Tile columns (B and C).
`BLOCK_K`	16	Tile depth: the K-dimension of each low-precision matrix multiply. Maps to a single SIMD register width for VNNI (128-bit at BLOCK_K=16, 256-bit at 32, 512-bit at 64) and similarly for BF16 (256-bit at BLOCK_K=16, 512-bit at 32, 2×512-bit at 64).
`BATCH_K`	4	Number of BLOCK_K panels grouped into one batch. The effective K-step per batch is BATCH_K x BLOCK_K. Larger values reduce barrier overhead and improve C-tile reuse at the cost of increased panel memory.

Example:

```
make ECFLAGS="-DBLOCK_K=32 -DBATCH_K=2" dgemm-wrap.x
```

Environment Variables

| Variable | Default | Description | ||::|| | `OZAKI` | 1 | Scheme selector: 0 = bypass (call original BLAS directly), 1 = Scheme 1 (mantissa slicing, int8), 2 = Scheme 2 (CRT), 3 = Scheme 3 (BF16 Dekker split), 4 = Scheme 4 (CRT + BF16, no OpenCL). | | `OZAKI_N` | *per scheme* | Number of decomposition units: slices for Scheme 1 (double: 1..16, default 8; float: 1..8, default 4) or moduli for Scheme 2 (see Scheme 2 section for per-precision defaults). | | `OZAKI_TM` | auto | Output tile height (multiple of 8). 0 = auto-select (start at 256, halve to fit work-group size). | | `OZAKI_TN` | auto | Output tile width (multiple of 16). 0 = auto-select (start at 256, halve to fit work-group size). | | `OZAKI_GROUPS` | 0 | Scheme 2 only: K-grouping factor (0/1 = disabled). When > 1, that many consecutive K sub-panels share one Garner reconstruction. | | `OZAKI_OCL` | 1 | Enable (1) or disable (0) the OpenCL/GPU path at runtime. Only effective when built with LIBXSTREAM support (`__LIBXSTREAM`). When disabled, falls back to the CPU Ozaki scheme. | | `OZAKI_FLAGS` | 3 | Scheme 1 bitmask: Triangular (1), Symmetrize (2); see above. | | `OZAKI_TRIM` | 0 | Scheme 1 diagonal trim: 0 = exact, T = drop T least significant diagonals (~7 bits each). | | `OZAKI_EPS` | inf | Dump A/B matrices as MHD-files when the epsilon error exceeds the given threshold (implies `OZAKI_VERBOSE=1` if unset). | | `OZAKI_VERBOSE` | 0 | 0 = silent; 1 = print accumulated statistic at exit; N = print every Nth GEMM call. | | `OZAKI_STAT` | 0 | Track C-matrix (0), A-matrix representation (1), or B-matrix representation (2). | | `OZAKI_EXIT` | 1 | Exit with failure after dumping matrices on accuracy violation (eps/rsq threshold exceeded). Set to 0 to continue execution. | | `OZAKI_RSQ` | 0 | Dump A/B matrices as MHD-files when RSQ drops below the given threshold; the threshold is updated after each dump (implies `OZAKI_VERBOSE=1` if unset). | | `CHECK` | 0 | Accuracy validation against BLAS reference: 0 = disabled, negative = auto-threshold (1e-10 for double, 1e-3 for float), positive = use value as threshold. Prints `CHECK: eps=...` to stderr and exits with failure if exceeded. | | `NREPEAT` | 1 | Number of GEMM calls; when > 1 the first call is warmup and excluded from timing. |

Test Driver

The test driver (`gemm.c`) accepts positional arguments:

```
dgemm-wrap.x [A.mhd|M [B.mhd|N] [K [TA [TB [ALPHA [BETA [LDA [LDB [LDC]]]]]]]]]
sgemm-wrap.x [A.mhd|M [B.mhd|N] [K [TA [TB [ALPHA [BETA [LDA [LDB [LDC]]]]]]]]]
dgemm-blas.x [A.mhd|M [B.mhd|N] [K [TA [TB [ALPHA [BETA [LDA [LDB [LDC]]]]]]]]]
sgemm-blas.x [A.mhd|M [B.mhd|N] [K [TA [TB [ALPHA [BETA [LDA [LDB [LDC]]]]]]]]]
```

TA and TB select transposition: 0 means 'N' (no transpose), non-zero means 'T' (transpose). The `dgemm-*` and `sgemm-*` drivers call DGEMM or SGEMM respectively, built with the matching `GEMM_REAL_TYPE` (`double` or `float`). The `GEMM_INT_TYPE` (default `int`) can also be overridden at build time.

Source Layout

| File | Purpose | ||| | `ozaki.h` | Shared header: block sizes (`BLOCK_M/BLOCK_N/BLOCK_K/BATCH_K`), slice and prime constants, IEEE-754 decomposition helpers, flag definitions (`OZ1_TRIANGULAR`, `OZ1_SYMMETRIZE`), and inline utility functions used by both schemes. | | `gemm.h` | Common header: type macros (`GEMM_ARGDECL/GEMM_ARGPASS`), precision-specific name redirects, function prototypes for all four GEMM flavors. | | `ozaki.c` | Wrapper/orchestration for real GEMM (`GEMM_WRAP`): initialization, environment handling, fallback dispatch, and global state management. Compiled twice (double + float). | | `ozaki1_int8.c` | Ozaki Scheme-1 computational kernel (`gemm_oz1`): decomposes IEEE-754 mantissa into 7-bit int8 slices for low-precision dot products. Uses function-pointer dispatch for VNNI vs scalar int8 dot product. Compiled twice (double + float). | | `ozaki2_int8.c` | Ozaki Scheme-2 computational kernel (`gemm_oz2`): CRT-based modular arithmetic using small pairwise coprime moduli. Barrett reduction for fast modular arithmetic, Garner's algorithm with batched reconstruction. Residues fit in int8 (sign folded in) and dot products use VNNI. Compiled twice (double + float). | | `ozaki1_bf16.c` | Ozaki Scheme-3 computational kernel (`gemm_oz3`): Dekker-style error-free split into BF16 slices each carries its own exponent, so no shared-exponent alignment is needed. Dot products via `VDPBF16PS` (scalar fallback). Inherently approximate (FP32 accumulation rounding). Compiled twice (double + float). | | `ozaki2_bf16.c` | Ozaki Scheme-4 computational kernel (`gemm_oz4`): CRT-based modular arithmetic (same decomposition as Scheme 2) with BF16 dot products via `VDPBF16PS` instead of VNNI int8. Residues (0--127) are exactly representable in BF16; FP32 accumulation is exact for BLOCK_K <= 64. Select with `OZAKI=4`. Compiled twice (double + float). | | `zgemm3m.c` | Complex GEMM 3M wrapper (`ZGEMM_WRAP`): deinterleaves complex matrices, issues 3 real GEMM calls (Karatsuba), recombines. Uses `libxs_malloc` for workspace. Compiled twice (double + float). | | `ozaki_ocl.c` | OpenCL bridge: wraps LIBXSTREAM behind an opaque handle (`ozaki_ocl_create/release/gemm/finalize`). Compiled only when LIBXSTREAM is detected; isolates all OpenCL includes from the rest of the code. | | `wrap.c` | Entry points (`GEMM`, `ZGEMM`) and dlsym fallbacks (`GEMM_REAL`, `ZGEMM_REAL`) via `GEMM_DEFINE_DLSYM` macro. Used only in the

LD_PRELOAD path; excluded from the static archive to keep `__real_` resolution correct. | | `gemm.c` | Test driver. Compiled as `dgemm-{wrap,blas}.x` (double) and `sgemm-{wrap,blas}.x` (float). | | `gemm-print.c` | `print_gemm` and `print_diff` utilities. | | `test-wrap.sh` | Integration test: exercises STATIC WRAP, ORIGINAL BLAS, and LD_PRELOAD paths. Auto-discovers built `*-wrap.x` / `*-blas.x` executables; accepts optional test-name prefix as first argument. | | `test-check.sh` | Correctness test: runs all four Ozaki schemes with `CHECK` validation. Auto-discovers built `*gemm-wrap.x` executables; accepts optional test-name prefix. | | `test-mhd.sh` | MHD file test: runs GEMM on all A/B MHD-file pairs in a directory. Accepts optional test-name prefix and directory arguments. |

If the driver is called with MHD-files, accuracy issues can be analyzed outside of an application. Microbenchmark

This code sample benchmarks the performance of the registry (key-value store) dispatch path. Various scenarios are measured to characterize the registry under different access patterns and concurrency levels.

Command Line Interface (CLI)

- Optionally takes the number of unique keys to register (default: 10000).
- Optionally takes the number of repeat iterations for lookup phases (default: 10).
- Optionally takes the number of threads (default: max available).

Measurements (Benchmark)

- Duration to register (insert) all keys into the registry (write path).
- Duration to look up keys with a shuffled access pattern that defeats the thread-local cache ("cold lookup").
- Duration to look up keys with a sequential/repeating pattern that hits the thread-local cache ("cached lookup").
- Duration of multi-threaded parallel reads across all threads.
- Duration of contended parallel writes (each thread registers its own key range).
- Duration of mixed read/write: one writer thread registering keys while remaining threads read concurrently.

In case of a multi-threaded benchmark, the timings represent contended requests. For thread-scaling, read-only accesses (lookup) stay roughly constant in duration per-op due to the thread-local cache, whereas write-accesses (registration) are serialized and duration scales with the number of threads.

Memory Allocation (Microbenchmark)

Benchmarks pooled scratch memory allocation (`libxs_malloc` / `libxs_free`) against the standard C library `malloc` / `free`. The pool-based allocator recycles memory after an initial warm-up, avoiding repeated system calls in steady state. The benchmark measures allocation throughput (calls/s) under a randomized mix of allocation sizes and counts, optionally with multiple threads. A Fortran variant compares `libxs_malloc` against Fortran `ALLOCATE` / `DEALLOCATE`.

Building

```
cd samples/scratch
make
```

Produces `scratch.x` (C) and, if a Fortran compiler is found, `scratchf.x`.

Usage (C)

```
./scratch.x [ncycles [max_nactive [nthreads]]]
```

| Argument | Default | Description | |||| | `ncycles` | 100 | Number of allocation/deallocation cycles | | `max_nactive` | 4 | Maximum number of concurrent live allocations per cycle (clamped to `MAX_MALLOC_N`, default 24) | | `nthreads` | 1 | OpenMP threads (clamped to `omp_get_max_threads()`) |

Environment Variables | Variable | Default | Description | |||| | `CHECK` | 0 | Non-zero: `memset` each allocation to detect mapping errors (slower, validates that pages are writable) |

Compile-time Knobs | Macro | Default | Description | |||| | `MAX_MALLOC_MB` | 100 | Upper bound on individual allocation size in MB (each allocation is 1--`MAX_MALLOC_MB` MB) | | `MAX_MALLOC_N` | 24 | Array size for the random-number table and per-cycle allocation slots |

```
## default: 100 cycles, up to 4 active allocations, 1 thread
./scratch.x

## heavy load: 500 cycles, up to 12 active allocations, 4 threads
./scratch.x 500 12 4

## validate pages are writable
CHECK=1 ./scratch.x 43 8 4
```

Usage (Fortran)

```
./scratchf.x [mbytes [nrepeat]]
```

| Argument | Default | Description | |||| | `mbytes` | 100 | Allocation size in MB | | `nrepeat` | 20 | Number of repetitions |

A simpler single-threaded benchmark that times a single `libxs_malloc` / `libxs_free` pair against Fortran `ALLOCATE` / `DEALLOCATE` per iteration. Reports per-iteration times (ms) and an average speedup ratio.

```
## default: 100 MB, 20 repetitions
./scratchf.x

## 256 MB, 50 repetitions
./scratchf.x 256 50
```

What Is Measured

Each cycle draws a random number of allocations (1 to `max_nactive`) with random sizes (1--`MAX_MALLOC_MB` MB each). The cycle allocates all buffers, optionally touches them (`CHECK`), then frees them in order.

1. **Standard malloc** `malloc(nbytes)` / `free(p)` (or TBB / Intel KMP variants depending on compile-time configuration). Runs first to avoid warm-page bias from the pool path.
2. **libxs scratch pool** `libxs_malloc(pool, nbytes, 0)` / `libxs_free(p)`. The pool grows on first use and recycles memory thereafter.

An untimed warm-up cycle runs both allocators before the measured loops to demand-page memory and stabilize CPU frequency. Both `malloc` and `free` (or their equivalents) are individually timed and included in the throughput calculation.

Both paths use the same randomized sequence and are timed with `libxs_timer_ncycles`. Reported metrics:

| Metric | Description | ||| | **calls/s (kHz)** | Allocation+free throughput for each allocator | | **Scratch size** | High-water mark of the pool (MB) | | **Malloc size** | Peak aggregate allocation per cycle (MB) | | **Scratch Speedup** | Ratio of pool throughput to stdlib throughput | | **Fair** | Size-adjusted speedup: `(malloc_size / scratch_size) * speedup`, accounting for the pool's memory overhead |

Shuffle

Benchmarks three shuffling strategies and compares their throughput:

| Label | Method | |:|:| | **RNG-shuffle** | Fisher--Yates shuffle using `libxs_rng_u32` for random indices and `LIBXS_MEMSWP` for element swaps (reference baseline). | | **DS1-shuffle** | `libxs_shuffle` -- deterministic in-place shuffle based on a coprime stride. | | **DS2-shuffle** | `libxs_shuffle2` -- deterministic out-of-place shuffle (source -> destination). |

Each iteration works on an array of unsigned integers initialized to `0 .. n-1`. On the final iteration the shuffled result can optionally be written as an MHD image file (one per method) or analyzed for randomness quality.

Build

```
cd samples/shuffle
make
```

Usage

```
./shuffle.x [nelems [elemsize [niters [repeat]]]]
```

| Positional | Default | Description | |:|:|:| | `nelems` | 64 MB / `elemsize` | Number of elements. | | `elemsize` | 4 | Size of each element in bytes (1, 2, 4, or 8). | | `niters` | 1 | Number of shuffle passes per timed measurement (averaged). | | `repeat` | 3 | Number of timed iterations; the first is treated as warm-up. |

Environment Variables | Variable | Default | Description | |:|:|:| | `RANDOM` | 0 (off) | If nonzero, count inversions (via merge-sort) of the shuffled data and report a randomness percentage (`rand=%`) instead of writing MHD files. | | `SPLIT` | 1 | Partitioning depth for the `STATS` metrics (imbalance and distance). | | `STATS` | 0 (off) | If nonzero, compute and print partition imbalance (`imb`) and Manhattan distance (`dst`) metrics per method. |

```
## Default run (64 MB of 4-byte elements, 3 repeats)
./shuffle.x

## 1M elements of 8 bytes, 4 shuffle passes, 5 repeats
./shuffle.x 1000000 8 4 5

## Enable randomness quality metric
RANDOM=1 ./shuffle.x 10000 4 1 3

## Enable partition statistics with split depth 2
STATS=1 SPLIT=2 ./shuffle.x
```

What Is Measured

Reported bandwidth is `2 x data-size / time` (in MB/s). The factor of two accounts for reading and writing each element during the shuffle. For the RNG (Fisher--Yates) method this is an approximation: elements where the random index equals the current index (~37%) are not swapped, so actual data movement is slightly lower. The first iteration is excluded from the arithmetic average.

Optional Quality Metrics

- `rand` -- Enabled by `RANDOM=1`. After shuffling, inversions are counted via merge-sort (O(n log n)) and compared against the expected count for a random permutation (n(n-1)/4) to give a percentage. For the stochastic RNG-shuffle, quality metrics are averaged across all non-warm-up iterations.
- `dst` -- Manhattan distance of the element sum from the expected uniform value, split into hierarchical partitions (`SPLIT`).
- `imb` -- Partition imbalance, measuring how unevenly element sums are distributed across sub-partitions (`SPLIT`).

MHD Image Output When `RANDOM` is not set and the element size matches a supported type (1, 2, 4, or 8 bytes), the final shuffled array is written as an MHD image:

| File | Source | |:|:| | `shuffle_rng.mhd` | RNG-shuffle result | | `shuffle_ds1.mhd` | DS1-shuffle result | | `shuffle_ds2.mhd` | DS2-shuffle result |

The images are shaped as close to square as possible (`isqrt(n) x n/isqrt(n)`) and converted to unsigned 8-bit via modulus, allowing visual inspection of shuffle uniformity.

Compile-Time Knobs

The Makefile inherits settings from `Makefile.inc`. Relevant flags:

| Variable | Default | Effect | |:|:|:| | `OMP` | 0 | Enable OpenMP (not used by this sample). | | `SYM` | 1 | Include debug symbols (`-g`). | | `BUBBLE_SORT` | *undefined* | Define (`-DBUBBLE_SORT`) to use O(n2) bubble-sort for the `RANDOM` metric instead of the default O(n log n) merge-sort inversion counter. |

Synchronization Primitives

Micro-benchmark for the lock implementations provided by LIBXS (`libxs_sync.h`). The program measures **single-thread latency** (uncontended acquire/release) and **multi-thread throughput** (mixed read/write workload) for every lock kind compiled into the library:

| Lock kind | Description | ||| | `LIBXS_LOCK_DEFAULT` | Compile-time default (typically `atomic`) | | `LIBXS_LOCK_SPINLOCK` | OS-native or CAS-based spin lock | | `LIBXS_LOCK_MUTEX` | OS-native mutex (`pthread_mutex_t`) | | `LIBXS_LOCK_RWLOCK` | Reader/writer lock (atomic or `pthread_rwlock_t`) |

Build

```
cd samples/sync
make GNU=1
```

OpenMP is enabled by default (`OMP=1`) for multi-threaded throughput tests.

Run

```
./sync.x [nthreads] [wratio%] [work_r] [work_w] [nlat] [ntpt]
```

| Argument | Default | Description | |||| | `nthreads` | all available | Number of OpenMP threads | | `wratio%` | 5 | Percentage of write operations (0-100) | | `work_r` | 100 | Simulated work inside read-critical section (cycles) | | `work_w` | 10 x work_r | Simulated work inside write-critical section (cycles) | | `nlat` | 2000000 | Number of iterations for latency measurement | | `ntpt` | 10000 | Number of iterations per thread for throughput measurement |

```
$ ./sync.x 4 5 100 1000
LIBXS: default lock-kind "atomic" (Other)

Latency and throughput of "atomic" (default) for nthreads=4 wratio=5% ...
        ro-latency: 11 ns (call/s 91 MHz, 33 cycles)
        rw-latency: 11 ns (call/s 90 MHz, 33 cycles)
        throughput: 0 us (call/s 9128 kHz, 328 cycles)
...
```

Measurement Details

- **RO-latency**: Uncontended read-lock acquire/release pairs (4x unrolled), reported as nanoseconds per operation and TSC cycles.
- **RW-latency**: Uncontended write-lock acquire/release pairs (4x unrolled).
- **Throughput**: All threads run a mixed read/write workload governed by `wratio%`. Simulated work inside the critical section is subtracted so only synchronization overhead is reported.